PORTFOLIO PROJECT · PYTHON · MACHINE LEARNING

# What drives profit, and can we predict it?

Cleaning, exploring and modeling two years of sales transactions with pandas and scikit-learn

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700 transactions · 2013–2014

DATA PREPARATION

# Five segments, five countries, sixteen fields

### Clean

Stripped dollar signs and dashes from the discount field. Blank discounts became zero.

### Treat outliers

Flagged extreme gross sales, sales and profit with the IQR rule and replaced them with the median.

### Engineer

Added profit margin, profit per unit and a four-level profit category. Parsed dates for trends.

WHAT THE DATA SHOWS

# Profit follows sales volume, not production cost

**0.48**

correlation between sales and profit

**0.02**

correlation between manufacturing price and profit

**43%**

of transactions are Government, the largest segment

Seasonality: profit peaks in June and December in both 2013 and 2014.

PREDICTING PROFIT

# A decision tree beat linear regression by 25%




Mean absolute error on the held-out test set. Lower is better. Features: units sold, gross sales, discounts, segment.

RECOMMENDATIONS

# Four moves for the business

### Plan for June and December

Build stock and time campaigns ahead of the two yearly profit peaks.

### Audit discounts against margin

Check each discount band's effect on profit margin, not just on sales.

### Tailor offers to Government

The largest segment by volume deserves its own pricing and account plan.

### Track margin per unit

Use the new profit-per-unit metric to find and fix weak products.

HONEST READ

# The best model still misses by about $1.1M, against an average profit near $1.5M

### Keep the accounting intact

Replacing outliers column by column breaks profit = sales − cost.

### Add product and price

Product and sale price set margin, and the model never saw them.

### Report a baseline

Compare every model against simply predicting the average.